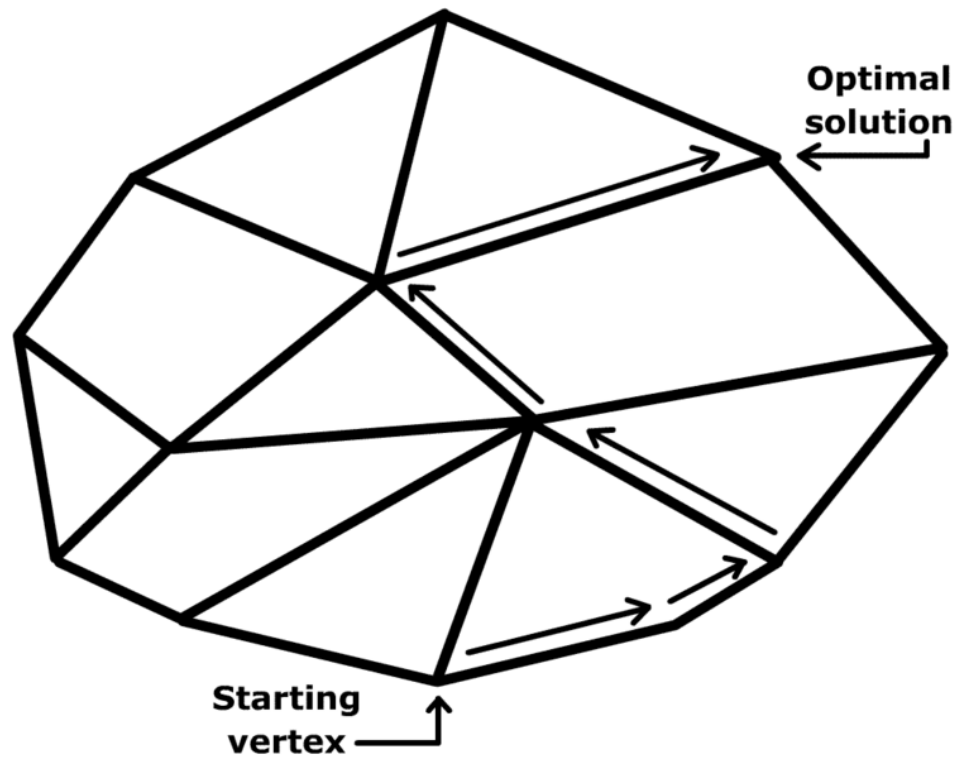


Algorithms and Analysis

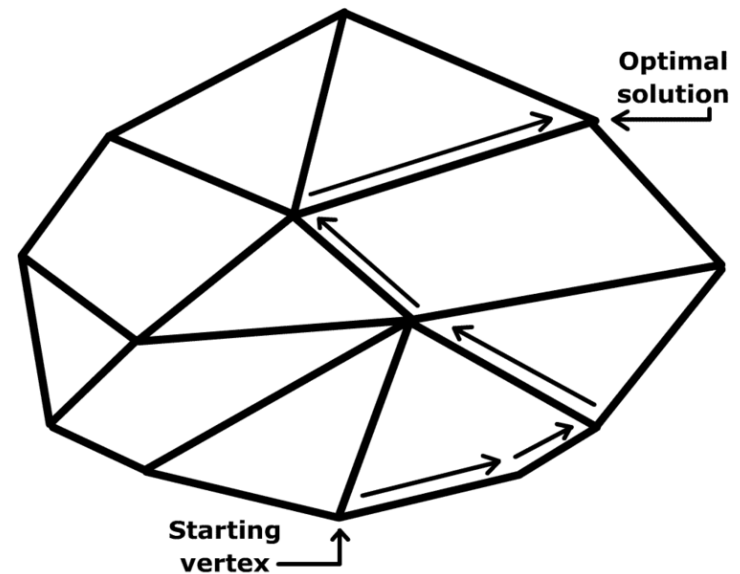
Lesson 28: *Solving Linear Programs*



linear programming, simplex methods, iterative search

Outline

1. **Recap**
2. Basic Feasible Solutions
3. Simplex Method
4. Classic LP Problems



Recap

- Linear programs are problems that can be formulated as follows

$$\min_{\mathbf{x}} \mathbf{c} \cdot \mathbf{x}$$

subject to

$$\mathbf{A}^{\leq} \mathbf{x} \leq \mathbf{b}^{\leq}, \quad \mathbf{A}^{\geq} \mathbf{x} \geq \mathbf{b}^{\geq}, \quad \mathbf{A}^{\text{=}} \mathbf{x} = \mathbf{b}^{\text{=}}, \quad \mathbf{x} \geq \mathbf{0}$$

- Where $\mathbf{x} = (x_1, x_2, \dots, x_n)$
- $\mathbf{A}^*$ are matrices and we interpret the inequalities to mean

$$\forall k \quad \sum_{j=1}^n A_{kj}^{\leq} x_j \leq b_k^{\leq}$$

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Optima and Vertices

- Because the objective function is linear ($c \cdot x$) there is a direction where the objective is always improving
- Thus, the optima cannot lie in the interior of the search space
- When we meet a constraint that limits the direction we can move, but we can still move along the constraint
- We then meet another constraint which restricts the direction we can move by two degrees of freedom
- Eventually, we will reach m constraints which defines a vertex of the feasible region and is optimal

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Transforming Linear Programs

- We can always transform an inequality constraint into an equality constraint by adding slack variables
- E.g.

$$a_1 \cdot x \geq 0 \quad \Rightarrow \quad a_1 \cdot x - z_1 = 0 \quad z_1 \geq 0$$

$$a_2 \cdot x \leq 0 \quad \Rightarrow \quad a_2 \cdot x + z_2 = 0 \quad z_2 \geq 0$$

z_1 (the excess) and z_2 (the deficit) are known as slack variables

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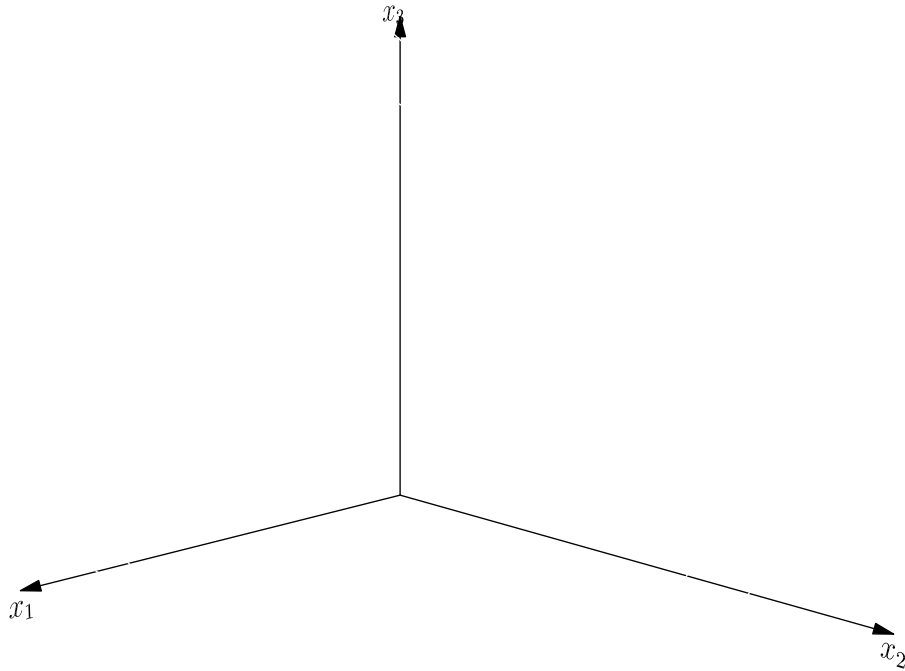
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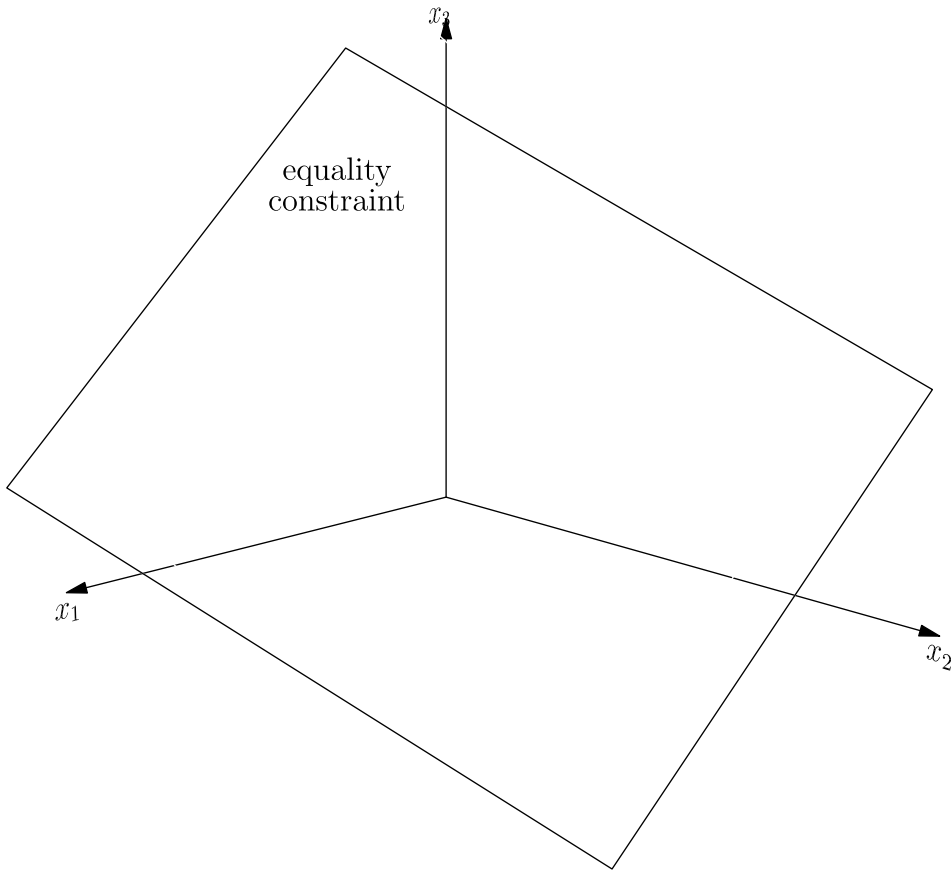
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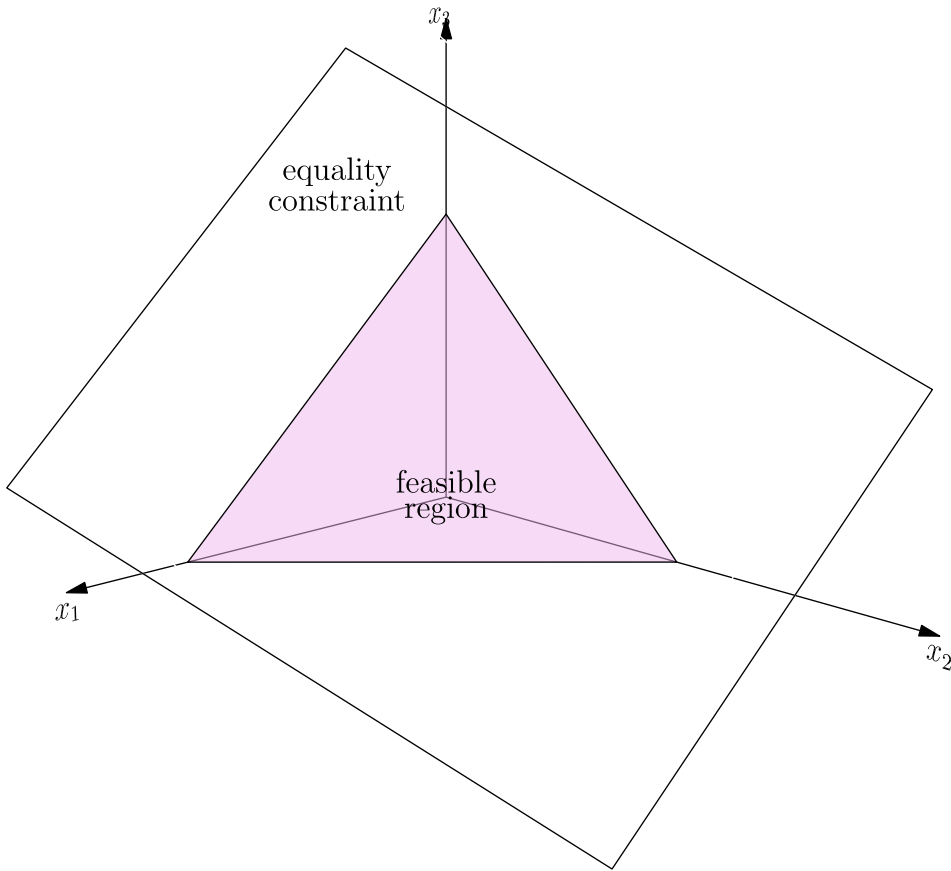
Solving Linear Programming



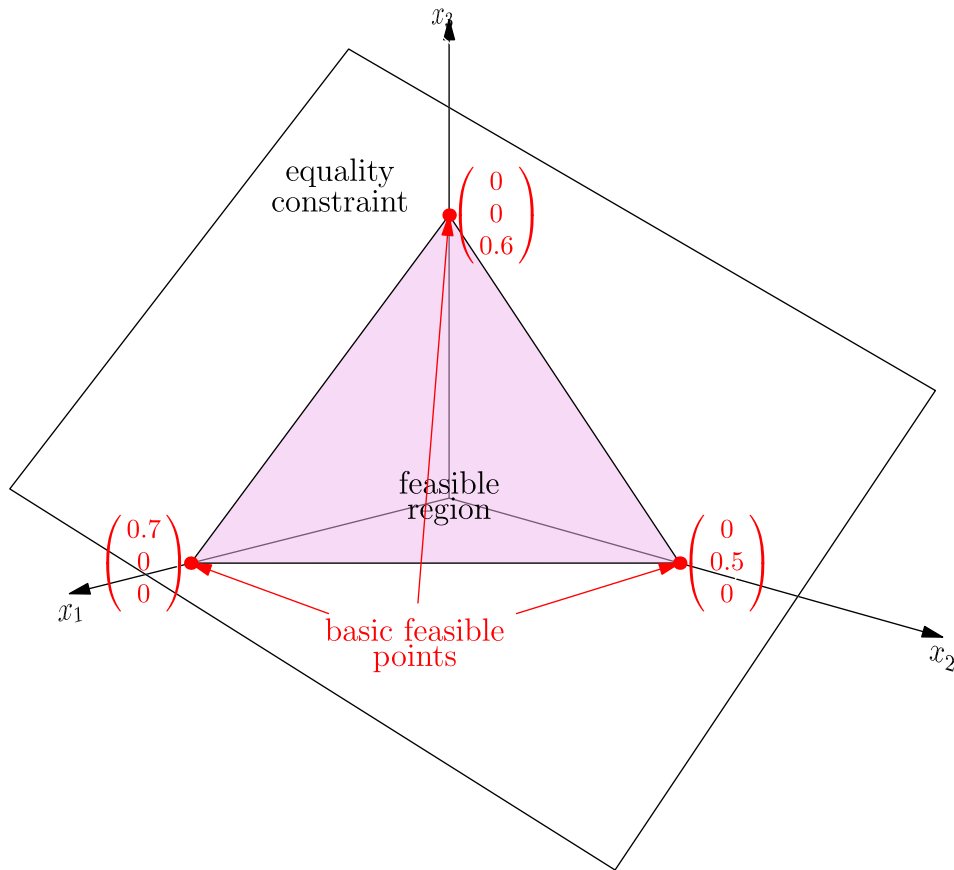
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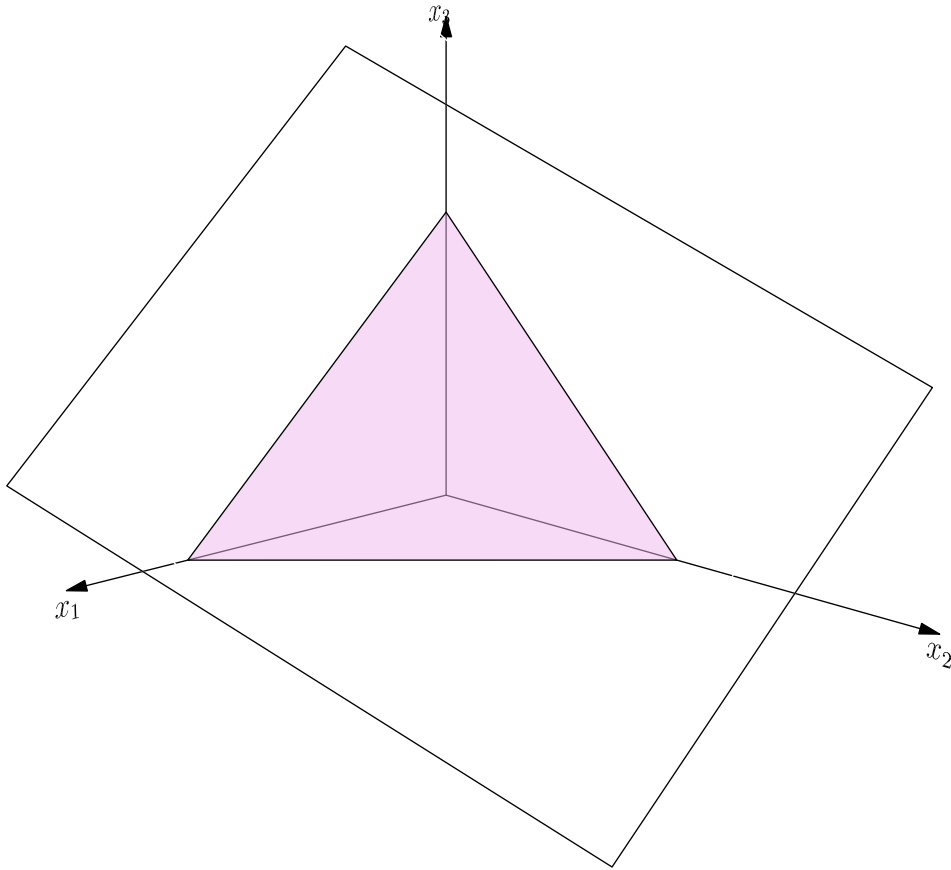
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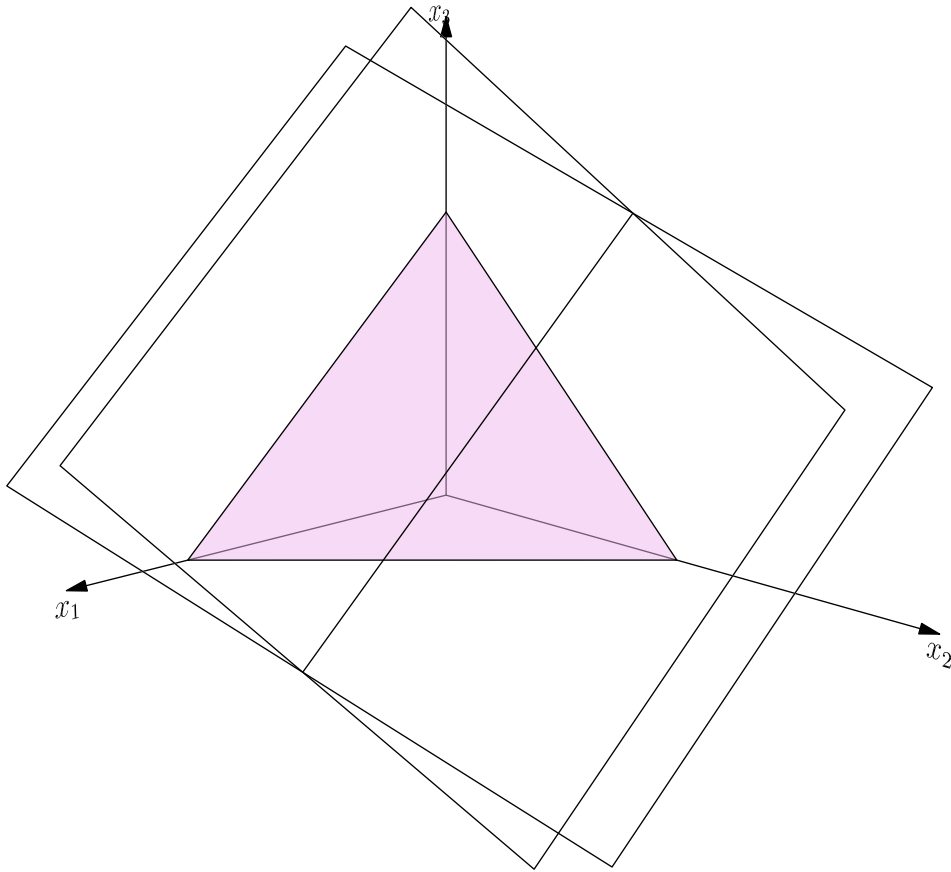
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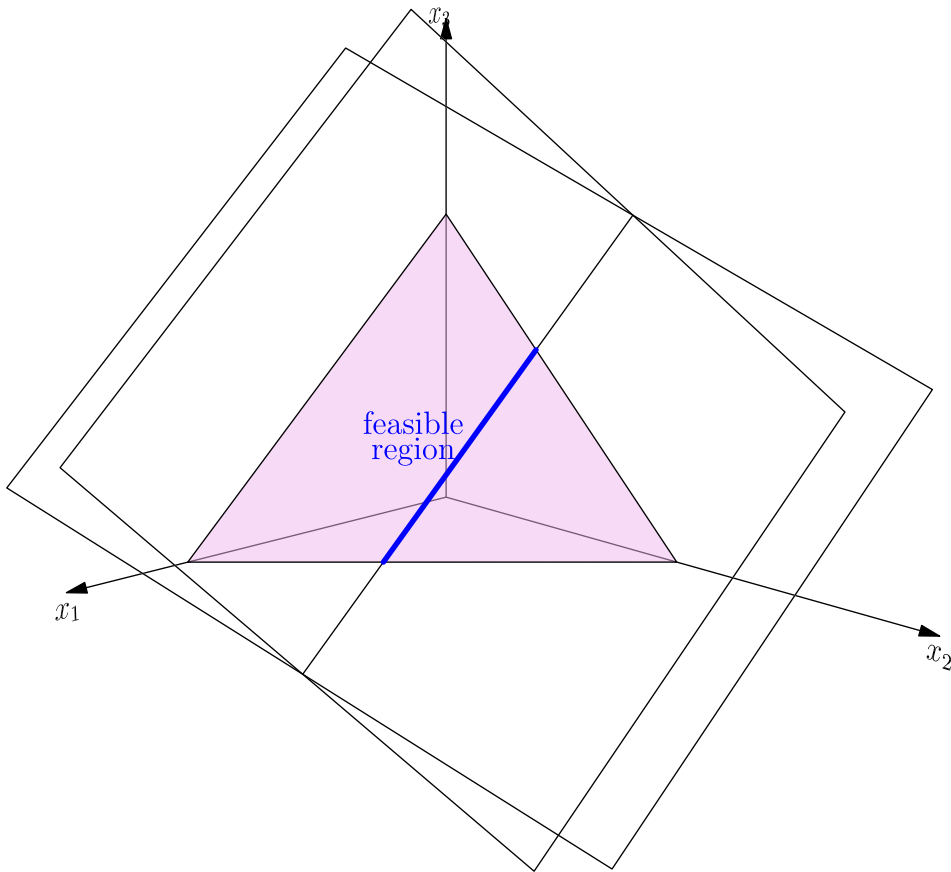
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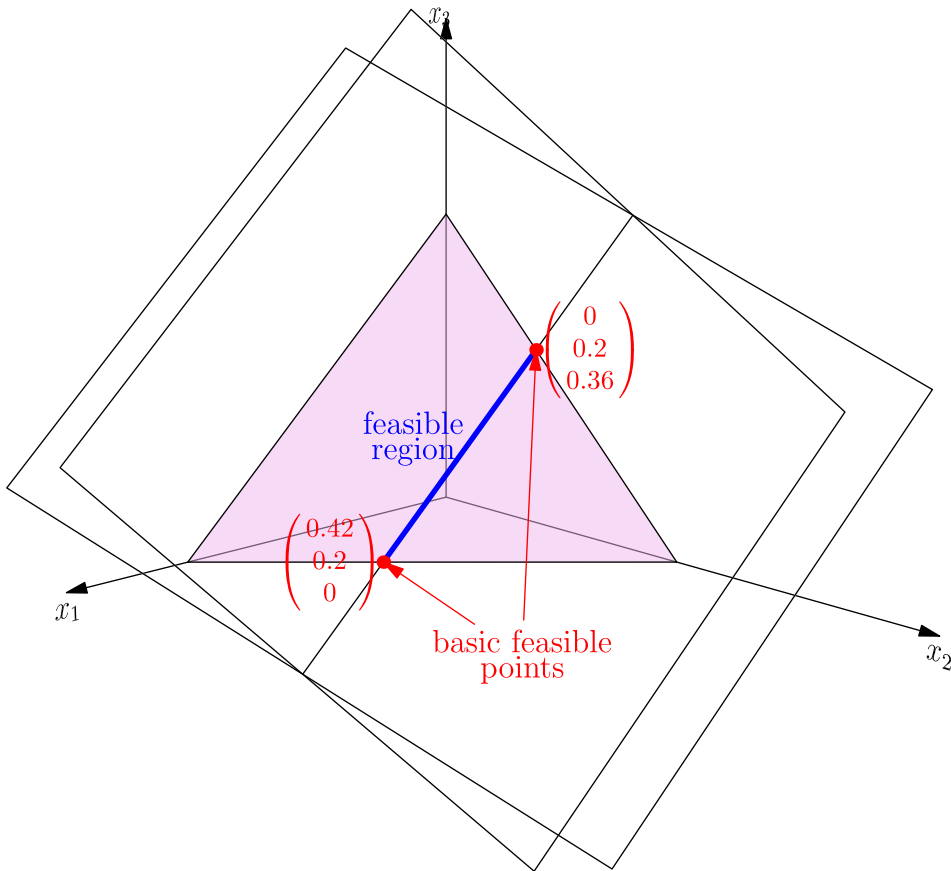
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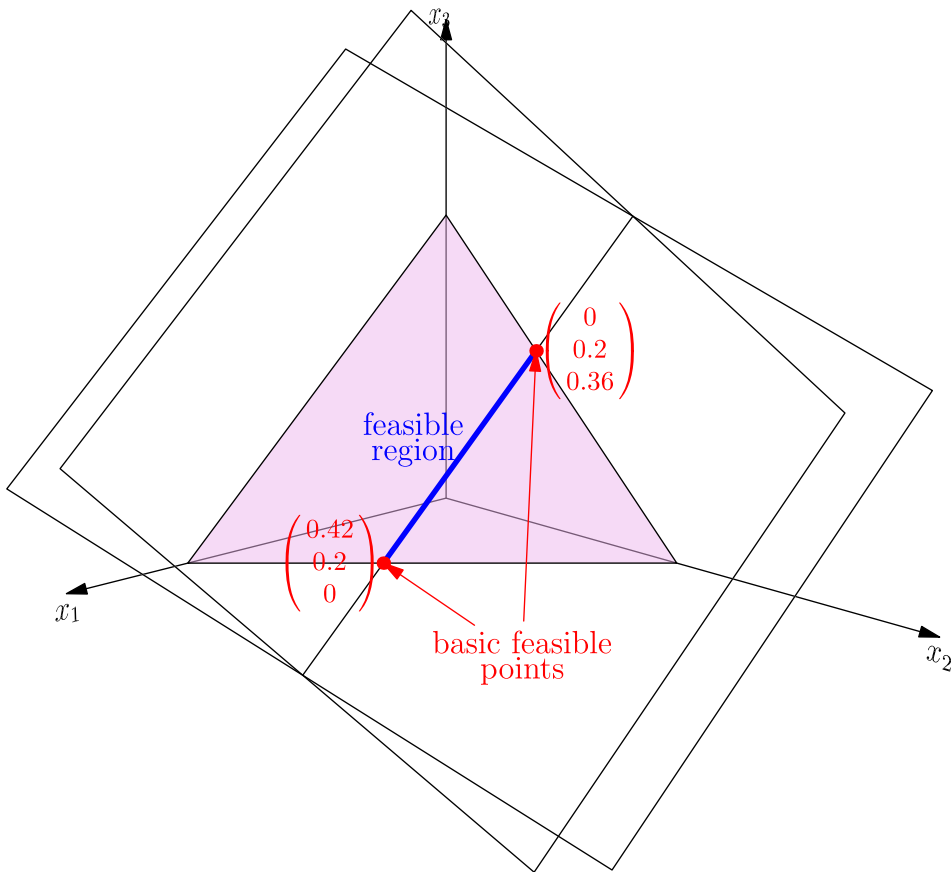


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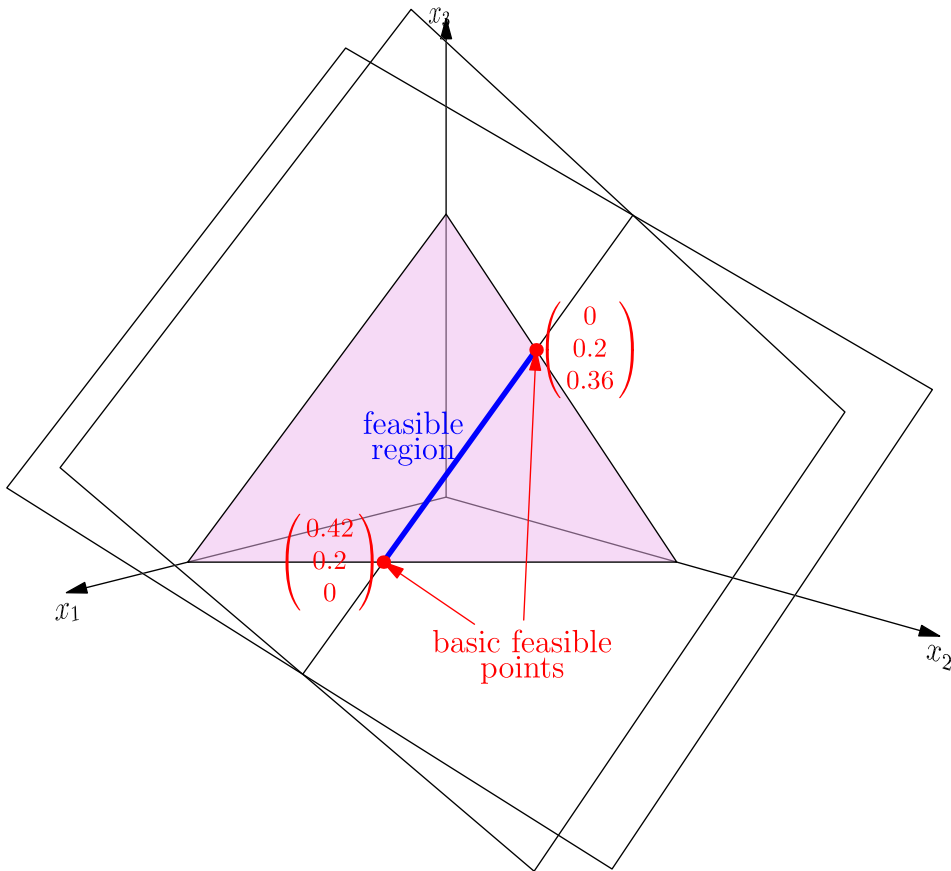


Solving Linear Programming

- The basic feasible points for LP problems with n variables and m constraints have at least $n - m$ zero variables

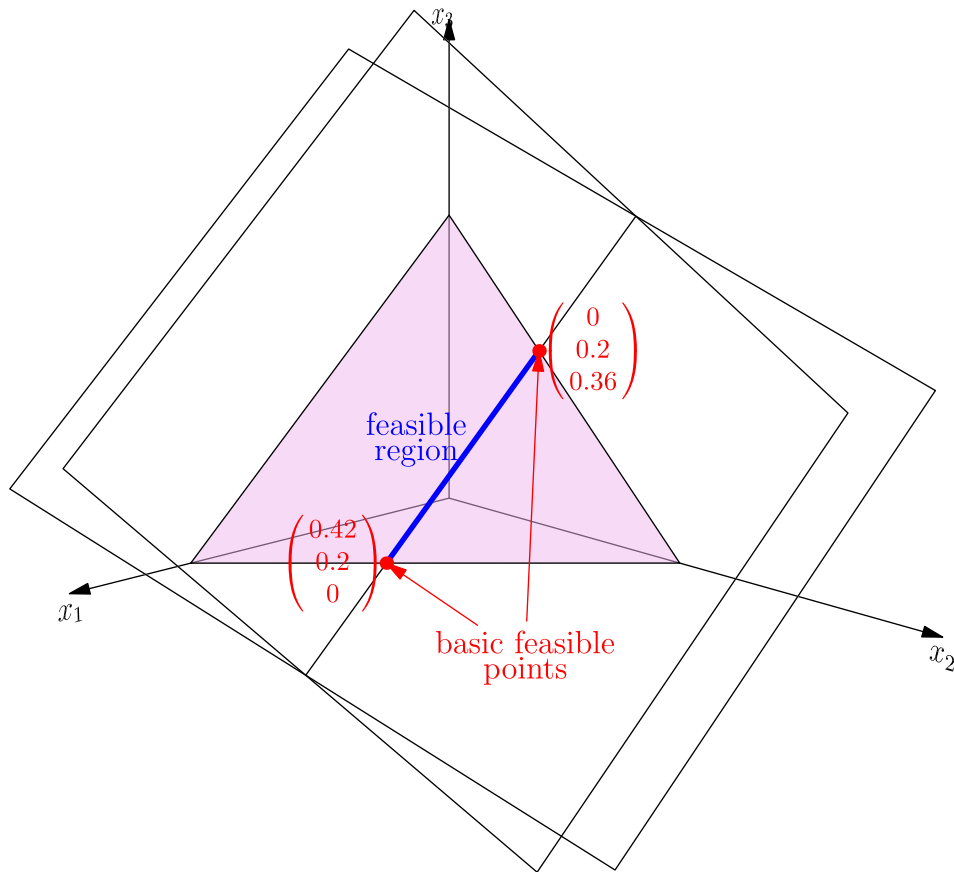


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- Typical number of basic feasible solutions is $\binom{n}{m} \geq \left(\frac{n}{m}\right)^m$

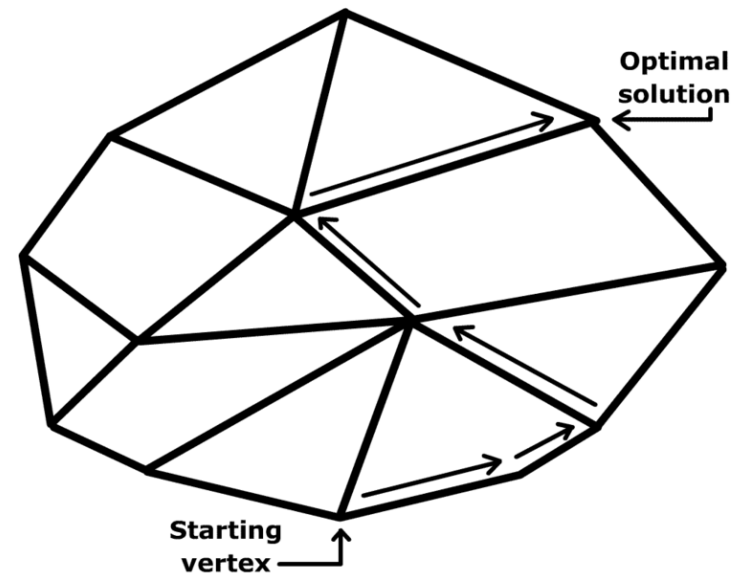
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- Typical number of basic feasible solutions is $\binom{n}{m} \geq \left(\frac{n}{m}\right)^m$
- Simplex algorithm organises iterative search for global solutions

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1. Recap
2. **Basic Feasible Solutions**
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Basic Feasible Solution

- *A basic feasible solution or basic feasible point is a solution that lies at a vertex of the feasible space*
- To solve a linear program we will start at a basic feasible point and move to the neighbour which best improves the objective function
- When we cannot find a better solution we are at the optimal solution
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Constraints

- There are two types of constraints
 1. n non-negativity constraints $x_i \geq 0$
 2. m additional constraints, which we can take to be equalities
$$\mathbf{A}x = b$$
- Note that some of the variables might be slack variables
- We consider the case when there are more variables than additional constraints, i.e. $n > m$
- This is usually be the case, but. . .
- If this isn't true it turns out you can consider an equivalent problem (dual problem) where you have a variable for each constraint and a constraint for each variable

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Basic Variable

- In total we have n equality and m non-negativity constraints
- m constraints must be satisfied to be at a vertex of feasible region
- So at least $n - m$ of the non-negativity constraints are satisfied (i.e. $x_i = 0$)
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- One of the tricky bits of tackling a linear program is to find an initial feasible solution
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- To do this for each additional constraint we add a new **auxiliary variable** ξ_k , e.g.

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- This should find a solution where all the $\xi_k = 0$
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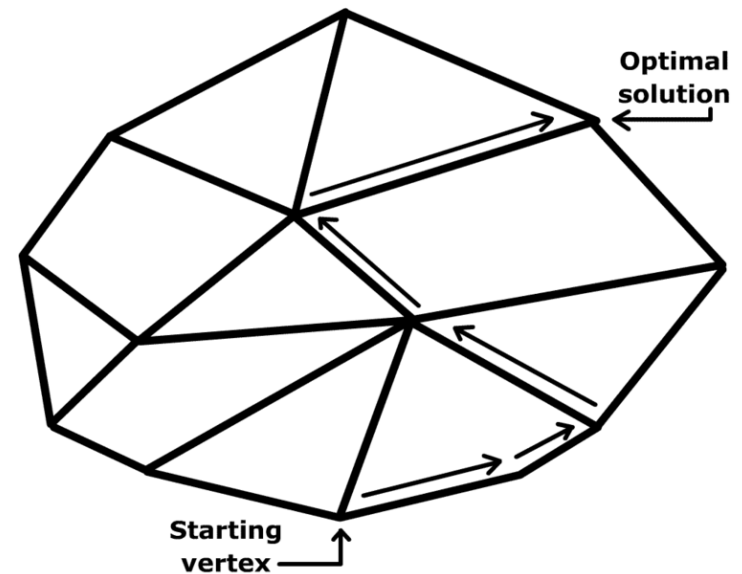
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- To perform the moves between vertices it helps to represent the problem in a **restricted normal form**
- Starting from a basic feasible point we have a constraint for each basic (non-zero) variable
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Tableau

$$\max_{\mathbf{x}} f(\mathbf{x}) = -1 - 7x_2 - 7x_3 + 5x_4 + 5x_5 + 8x_6 + 3x_7$$

where

$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7 \geq 0$$

$$x_8 = 3 + 6x_2 - 6x_3 + 2x_4 - 7x_5 + 2x_6 + 7x_7 \geq 0$$

$$x_9 = 5 - 3x_2 - x_3 - 2x_4 - 7x_5 - 7x_6 - 4x_7 \geq 0$$

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x_1	7	-1	3	-4	7	6	7
x_8	3	6	-6	2	-7	2	7
x_9	5	-3	-1	-2	-7	-7	-4

Tableau

$$\max_{\mathbf{x}} f(\mathbf{x}) = -1 - 7x_2 - 7x_3 + 5x_4 + 5x_5 + 8x_6 + 3x_7$$

where

$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7 \geq 0$$

$$x_8 = 3 + 6x_2 - 6x_3 + 2x_4 - 7x_5 + 2x_6 + 7x_7 \geq 0$$

$$x_9 = 5 - 3x_2 - x_3 - 2x_4 - 7x_5 - 7x_6 - 4x_7 \geq 0$$

$$x_2 = x_3 = x_4 = x_5 = x_6 = x_7 = 0$$

		x_2	x_3	x_4	x_5	x_6	x_7
$f(\mathbf{x})$	-1	-7	-7	5	5	8	3
x_1	7	-1	3	-4	7	6	7
x_8	3	6	-6	2	-7	2	7
x_9	5	-3	-1	-2	-7	-7	-4

Tableau

$$\max_{\mathbf{x}} f(\mathbf{x}) = -1 - 7x_2 - 7x_3 + 5x_4 + 5x_5 + 8x_6 + 3x_7$$

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$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7 \geq 0$$

$$x_8 = 3 + 6x_2 - 6x_3 + 2x_4 - 7x_5 + 2x_6 + 7x_7 \geq 0$$

$$x_9 = 5 - 3x_2 - x_3 - 2x_4 - 7x_5 - 7x_6 - 4x_7 \geq 0$$

$$x_2 = x_3 = x_4 = x_5 = x_6 = x_7 = 0$$

$$x_4 = 1.8, \quad x_1 = 0, \quad f(\mathbf{x}) = 7.8$$

		x_2	x_3	x_4	x_5	x_6	x_7
$f(\mathbf{x})$	-1	-7	-7	5	5	8	3
x_1	7	-1	3	-4	7	6	7
x_8	3	6	-6	2	-7	2	7
x_9	5	-3	-1	-2	-7	-7	-4

Tableau

$$\max_{\mathbf{x}} f(\mathbf{x}) = -1 - 7x_2 - 7x_3 + 5x_4 + 5x_5 + 8x_6 + 3x_7$$

where

$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7 \geq 0$$

$$x_8 = 3 + 6x_2 - 6x_3 + 2x_4 - 7x_5 + 2x_6 + 7x_7 \geq 0$$

$$x_9 = 5 - 3x_2 - x_3 - 2x_4 - 7x_5 - 7x_6 - 4x_7 \geq 0$$

$$x_2 = x_3 = x_4 = x_5 = x_6 = x_7 = 0$$

		x_2	x_3	x_4	x_5	x_6	x_7
$f(\mathbf{x})$	-1	-7	-7	5	5	8	3
x_1	7	-1	3	-4	7	6	7
x_8	3	6	-6	2	-7	2	7
x_9	5	-3	-1	-2	-7	-7	-4

Tableau

$$\max_{\mathbf{x}} f(\mathbf{x}) = -1 - 7x_2 - 7x_3 + 5x_4 + 5x_5 + 8x_6 + 3x_7$$

where

$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7 \geq 0$$

$$x_8 = 3 + 6x_2 - 6x_3 + 2x_4 - 7x_5 + 2x_6 + 7x_7 \geq 0$$

$$x_9 = 5 - 3x_2 - x_3 - 2x_4 - 7x_5 - 7x_6 - 4x_7 \geq 0$$

$$x_2 = x_3 = x_4 = x_5 = x_6 = x_7 = 0$$

$$x_5 = 0.43, \quad x_8 = 0, \quad f(\mathbf{x}) = 1.1$$

		x_2	x_3	x_4	x_5	x_6	x_7
$f(\mathbf{x})$	-1	-7	-7	5	5	8	3
x_1	7	-1	3	-4	7	6	7
x_8	3	6	-6	2	-7	2	7
x_9	5	-3	-1	-2	-7	-7	-4

Tableau

$$\max_{\mathbf{x}} f(\mathbf{x}) = -1 - 7x_2 - 7x_3 + 5x_4 + 5x_5 + 8x_6 + 3x_7$$

where

$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7 \geq 0$$

$$x_8 = 3 + 6x_2 - 6x_3 + 2x_4 - 7x_5 + 2x_6 + 7x_7 \geq 0$$

$$x_9 = 5 - 3x_2 - x_3 - 2x_4 - 7x_5 - 7x_6 - 4x_7 \geq 0$$

$$x_2 = x_3 = x_4 = x_5 = x_6 = x_7 = 0$$

		x_2	x_3	x_4	x_5	x_6	x_7
$f(\mathbf{x})$	-1	-7	-7	5	5	8	3
x_1	7	-1	3	-4	7	6	7
x_8	3	6	-6	2	-7	2	7
x_9	5	-3	-1	-2	-7	-7	-4

Tableau

$$\max_{\mathbf{x}} f(\mathbf{x}) = -1 - 7x_2 - 7x_3 + 5x_4 + 5x_5 + 8x_6 + 3x_7$$

where

$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7 \geq 0$$

$$x_8 = 3 + 6x_2 - 6x_3 + 2x_4 - 7x_5 + 2x_6 + 7x_7 \geq 0$$

$$x_9 = 5 - 3x_2 - x_3 - 2x_4 - 7x_5 - 7x_6 - 4x_7 \geq 0$$

$$x_2 = x_3 = x_4 = x_5 = x_6 = x_7 = 0$$

$$x_6 = 0.71, \quad x_9 = 0, \quad f(\mathbf{x}) = 4.7$$

		x_2	x_3	x_4	x_5	x_6	x_7
$f(\mathbf{x})$	-1	-7	-7	5	5	8	3
x_1	7	-1	3	-4	7	6	7
x_8	3	6	-6	2	-7	2	7
x_9	5	-3	-1	-2	-7	-7	-4

Tableau

$$\max_{\mathbf{x}} f(\mathbf{x}) = -1 - 7x_2 - 7x_3 + 5x_4 + 5x_5 + 8x_6 + 3x_7$$

where

$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7 \geq 0$$

$$x_8 = 3 + 6x_2 - 6x_3 + 2x_4 - 7x_5 + 2x_6 + 7x_7 \geq 0$$

$$x_9 = 5 - 3x_2 - x_3 - 2x_4 - 7x_5 - 7x_6 - 4x_7 \geq 0$$

$$x_2 = x_3 = x_4 = x_5 = x_6 = x_7 = 0$$

		x_2	x_3	x_4	x_5	x_6	x_7
$f(\mathbf{x})$	-1	-7	-7	5	5	8	3
x_1	7	-1	3	-4	7	6	7
x_8	3	6	-6	2	-7	2	7
x_9	5	-3	-1	-2	-7	-7	-4

Tableau

$$\max_{\mathbf{x}} f(\mathbf{x}) = -1 - 7x_2 - 7x_3 + 5x_4 + 5x_5 + 8x_6 + 3x_7$$

where

$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7 \geq 0$$

$$x_8 = 3 + 6x_2 - 6x_3 + 2x_4 - 7x_5 + 2x_6 + 7x_7 \geq 0$$

$$x_9 = 5 - 3x_2 - x_3 - 2x_4 - 7x_5 - 7x_6 - 4x_7 \geq 0$$

$$x_2 = x_3 = x_4 = x_5 = x_6 = x_7 = 0$$

$$x_7 = 1.2, \quad x_9 = 0, \quad f(\mathbf{x}) = 2.8$$

		x_2	x_3	x_4	x_5	x_6	x_7
$f(\mathbf{x})$	-1	-7	-7	5	5	8	3
x_1	7	-1	3	-4	7	6	7
x_8	3	6	-6	2	-7	2	7
x_9	5	-3	-1	-2	-7	-7	-4

Tableau

$$\max_{\mathbf{x}} f(\mathbf{x}) = -1 - 7x_2 - 7x_3 + 5x_4 + 5x_5 + 8x_6 + 3x_7$$

where

$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7 \geq 0$$

$$x_8 = 3 + 6x_2 - 6x_3 + 2x_4 - 7x_5 + 2x_6 + 7x_7 \geq 0$$

$$x_9 = 5 - 3x_2 - x_3 - 2x_4 - 7x_5 - 7x_6 - 4x_7 \geq 0$$

$$x_2 = x_3 = x_4 = x_5 = x_6 = x_7 = 0$$

Best pivot: $x_4 = 1.8$, $x_1 = 0$, $f(\mathbf{x}) = 7.8$ swap x_1 with x_4

		x_2	x_3	x_4	x_5	x_6	x_7
$f(\mathbf{x})$	-1	-7	-7	5	5	8	3
x_1	7	-1	3	-4	7	6	7
x_8	3	6	-6	2	-7	2	7
x_9	5	-3	-1	-2	-7	-7	-4

Tableau

Swap x_1 with x_4

$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7$$

$$\Rightarrow x_4 = 1.8 - 0.25x_2 + 0.75x_3 - 0.25x_1 + 1.8x_5 + 1.5x_6 + 1.8x_7$$

Tableau

Swap x_1 with x_4

$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7$$

$$\Rightarrow x_4 = 1.8 - 0.25x_2 + 0.75x_3 - 0.25x_1 + 1.8x_5 + 1.5x_6 + 1.8x_7$$

$$\max_{\mathbf{x}} f(\mathbf{x}) = -1 - 7x_2 - 7x_3 + 5x_4 + 5x_5 + 8x_6 + 3x_7$$

$$\Rightarrow \max_{\mathbf{x}} f(\mathbf{x}) = 7.8 - 8.2x_2 - 3.2x_3 - 1.2x_1 + 14x_5 + 16x_6 + 12x_7$$

Tableau

Swap x_1 with x_4

$$x_1 = 7 - x_2 + 3x_3 - 4x_4 + 7x_5 + 6x_6 + 7x_7$$

$$\Rightarrow x_4 = 1.8 - 0.25x_2 + 0.75x_3 - 0.25x_1 + 1.8x_5 + 1.5x_6 + 1.8x_7$$

$$\max_{\mathbf{x}} f(\mathbf{x}) = -1 - 7x_2 - 7x_3 + 5x_4 + 5x_5 + 8x_6 + 3x_7$$

$$\Rightarrow \max_{\mathbf{x}} f(\mathbf{x}) = 7.8 - 8.2x_2 - 3.2x_3 - 1.2x_1 + 14x_5 + 16x_6 + 12x_7$$

$$x_8 = 3 + 6x_2 - 6x_3 + 2x_4 - 7x_5 + 2x_6 + 7x_7$$

$$\Rightarrow x_8 = 6.5 + 5.5x_2 - 4.5x_3 - 0.5x_1 - 3.5x_5 + 5x_6 + 10x_7$$

$$x_9 = 5 - 3x_2 - x_3 - 2x_4 - 7x_5 - 7x_6 - 4x_7$$

$$\Rightarrow x_9 = 1.5 - 2.5x_2 - 2.5x_3 + 0.5x_1 - 10x_5 - 10x_6 - 7.5x_7$$

Tableau

$$\max_{\mathbf{x}} f(\mathbf{x}) = 7.8 - 8.2x_2 - 3.2x_3 - 1.2x_1 + 14x_5 + 16x_6 + 12x_7$$

where

$$x_4 = 1.8 - 0.25x_2 + 0.75x_3 - 0.25x_1 + 1.8x_5 + 1.5x_6 + 1.8x_7 \geq 0$$

$$x_8 = 6.5 + 5.5x_2 - 4.5x_3 - 0.5x_1 - 3.5x_5 + 5x_6 + 10x_7 \geq 0$$

$$x_9 = 1.5 - 2.5x_2 - 2.5x_3 + 0.5x_1 - 10x_5 - 10x_6 - 7.5x_7 \geq 0$$

$$x_2 = x_3 = x_1 = x_5 = x_6 = x_7 = 0$$

		x_2	x_3	x_1	x_5	x_6	x_7
$f(\mathbf{x})$	7.8	-8.2	-3.2	-1.2	14	16	12
x_4	1.8	-0.25	0.75	-0.25	1.8	1.5	1.8
x_8	6.5	5.5	-4.5	-0.5	-3.5	5	10
x_9	1.5	-2.5	-2.5	0.5	-10	-10	-7.5

Tableau

$$\max_{\mathbf{x}} f(\mathbf{x}) = 7.8 - 8.2x_2 - 3.2x_3 - 1.2x_1 + 14x_5 + 16x_6 + 12x_7$$

where

$$x_4 = 1.8 - 0.25x_2 + 0.75x_3 - 0.25x_1 + 1.8x_5 + 1.5x_6 + 1.8x_7 \geq 0$$

$$x_8 = 6.5 + 5.5x_2 - 4.5x_3 - 0.5x_1 - 3.5x_5 + 5x_6 + 10x_7 \geq 0$$

$$x_9 = 1.5 - 2.5x_2 - 2.5x_3 + 0.5x_1 - 10x_5 - 10x_6 - 7.5x_7 \geq 0$$

$$x_2 = x_3 = x_1 = x_5 = x_6 = x_7 = 0$$

Best pivot: $x_7 = 0.2$, $x_9 = 0$, $f(\mathbf{x}) = 10$ swap x_9 with x_7

		x_2	x_3	x_1	x_5	x_6	x_7
$f(\mathbf{x})$	7.8	-8.2	-3.2	-1.2	14	16	12
x_4	1.8	-0.25	0.75	-0.25	1.8	1.5	1.8
x_8	6.5	5.5	-4.5	-0.5	-3.5	5	10
x_9	1.5	-2.5	-2.5	0.5	-10	-10	-7.5

Tableau

Swap x_9 with x_7

$$x_9 = 1.5 - 2.5x_2 - 2.5x_3 + 0.5x_1 - 10x_5 - 10x_6 - 7.5x_7$$

$$\Rightarrow x_7 = 0.2 - 0.33x_2 - 0.33x_3 + 0.067x_1 - 1.4x_5 - 1.3x_6 - 0.13x_9$$

Tableau

Swap x_9 with x_7

$$x_9 = 1.5 - 2.5x_2 - 2.5x_3 + 0.5x_1 - 10x_5 - 10x_6 - 7.5x_7$$

$$\Rightarrow x_7 = 0.2 - 0.33x_2 - 0.33x_3 + 0.067x_1 - 1.4x_5 - 1.3x_6 - 0.13x_9$$

$$\max_{\mathbf{x}} f(\mathbf{x}) = 7.8 - 8.2x_2 - 3.2x_3 - 1.2x_1 + 14x_5 + 16x_6 + 12x_7$$

$$\Rightarrow \max_{\mathbf{x}} f(\mathbf{x}) = 10 - 12x_2 - 7.2x_3 - 0.47x_1 - 2.7x_5 - 0.17x_6 - 1.6x_9$$

Tableau

Swap x_9 with x_7

$$x_9 = 1.5 - 2.5x_2 - 2.5x_3 + 0.5x_1 - 10x_5 - 10x_6 - 7.5x_7$$

$$\Rightarrow x_7 = 0.2 - 0.33x_2 - 0.33x_3 + 0.067x_1 - 1.4x_5 - 1.3x_6 - 0.13x_9$$

$$\max_{\mathbf{x}} f(\mathbf{x}) = 7.8 - 8.2x_2 - 3.2x_3 - 1.2x_1 + 14x_5 + 16x_6 + 12x_7$$

$$\Rightarrow \max_{\mathbf{x}} f(\mathbf{x}) = 10 - 12x_2 - 7.2x_3 - 0.47x_1 - 2.7x_5 - 0.17x_6 - 1.6x_9$$

$$x_4 = 1.8 - 0.25x_2 + 0.75x_3 - 0.25x_1 + 1.8x_5 + 1.5x_6 + 1.8x_7$$

$$\Rightarrow x_4 = 2.1 - 0.83x_2 + 0.17x_3 - 0.13x_1 - 0.7x_5 - 0.83x_6 - 0.23x_9$$

$$x_8 = 6.5 + 5.5x_2 - 4.5x_3 - 0.5x_1 - 3.5x_5 + 5x_6 + 10x_7$$

$$\Rightarrow x_8 = 8.6 + 2x_2 - 8x_3 + 0.2x_1 - 18x_5 - 9x_6 - 1.4x_9$$

Tableau

$$\max_{\mathbf{x}} f(\mathbf{x}) = 10 - 12x_2 - 7.2x_3 - 0.47x_1 - 2.7x_5 - 0.17x_6 - 1.6x_9$$

where

$$x_4 = 2.1 - 0.83x_2 + 0.17x_3 - 0.13x_1 - 0.7x_5 - 0.83x_6 - 0.23x_9 \geq 0$$

$$x_8 = 8.6 + 2x_2 - 8x_3 + 0.2x_1 - 18x_5 - 9x_6 - 1.4x_9 \geq 0$$

$$x_7 = 0.2 - 0.33x_2 - 0.33x_3 + 0.067x_1 - 1.4x_5 - 1.3x_6 - 0.13x_9 \geq 0$$

$$x_2 = x_3 = x_1 = x_5 = x_6 = x_9 = 0$$

		x_2	x_3	x_1	x_5	x_6	x_9
$f(\mathbf{x})$	10	-12	-7.2	-0.47	-2.7	-0.17	-1.6
x_4	2.1	-0.83	0.17	-0.13	-0.7	-0.83	-0.23
x_8	8.6	2	-8	0.2	-18	-9	-1.4
x_7	0.2	-0.33	-0.33	0.067	-1.4	-1.3	-0.13

Tableau

$$\max_{\mathbf{x}} f(\mathbf{x}) = 10 - 12x_2 - 7.2x_3 - 0.47x_1 - 2.7x_5 - 0.17x_6 - 1.6x_9$$

where

$$x_4 = 2.1 - 0.83x_2 + 0.17x_3 - 0.13x_1 - 0.7x_5 - 0.83x_6 - 0.23x_9 \geq 0$$

$$x_8 = 8.6 + 2x_2 - 8x_3 + 0.2x_1 - 18x_5 - 9x_6 - 1.4x_9 \geq 0$$

$$x_7 = 0.2 - 0.33x_2 - 0.33x_3 + 0.067x_1 - 1.4x_5 - 1.3x_6 - 0.13x_9 \geq 0$$

$$x_2 = x_3 = x_1 = x_5 = x_6 = x_9 = 0$$

Optimum

		x_2	x_3	x_1	x_5	x_6	x_9
$f(\mathbf{x})$	10	-12	-7.2	-0.47	-2.7	-0.17	-1.6
x_4	2.1	-0.83	0.17	-0.13	-0.7	-0.83	-0.23
x_8	8.6	2	-8	0.2	-18	-9	-1.4
x_7	0.2	-0.33	-0.33	0.067	-1.4	-1.3	-0.13

Awkward Problems

- If there are any column with all entries positive then this variable can be increase forever—this is a signal that the linear programming problem is unbounded
- You can also find that a basic variable becomes zero—this is known as a degenerate feasible vector
- It can be removed by exchanging variables on the left of the inequality with variables on the right
- This makes the algorithm a bit more complex to implement

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High Performance Solvers

- Although the tableau method is the “classic solver” it doesn’t cut the mustard for large scale problems
- The simplex update can also be viewed as solving a linear set of equations which is facilitated by performing an LU-decomposition
- However, the constraints are often very sparse so good solvers try to take advantage of the sparsity
- Top end simplex algorithms are rather complex
- There is a second approach known as the interior point method which is competitive on large problems

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Time Complexity of Simplex

- The time complexity of the updates is $O(n^2)$
- The critical question is how many updates are necessary
- It turns out that typically this is $O(n)$ making the simplex algorithm $O(n^3)$
- However, it is possible to cook up problems where there is a “long path” from the initial solution to the optimum which is exponentially big
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Interior Point Method

- An alternative to the simplex method is the interior point method which always remains in the feasible region, away from the constraints
- These method iterate towards the constraints and are provably polynomial
- For small linear programming problems they are out-performed in practice by the simplex method
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Interior Point Method

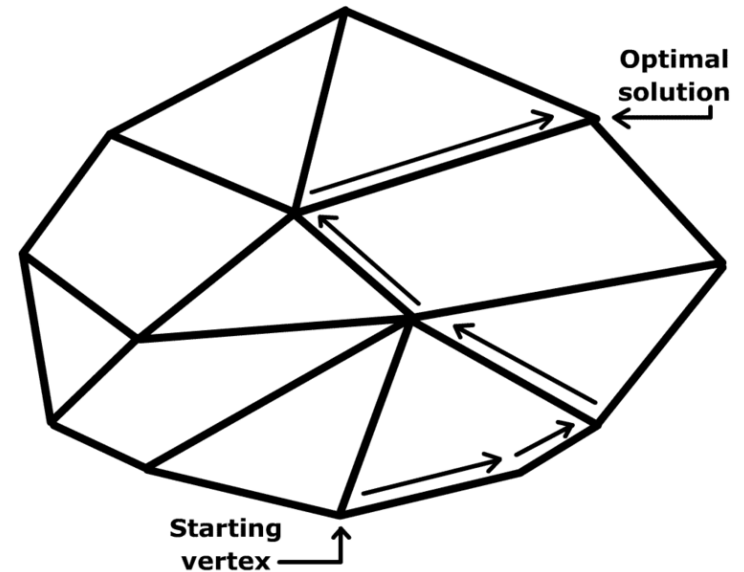
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Outline

1. Recap
2. Basic Feasible Solutions
3. Simplex Method
4. **Classic LP Problems**



LP Problems

- Any problem that can be set up as a linear program can be solved in polynomial time
- One way is just to feed it to a LP-solver
- Sometimes the problems are important enough and have such a distinctive formulation that faster specialised algorithms have been developed
- We consider a couple of classic problems: *maximum flow* and *linear assignment*

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Maximum Flow

- In maximum flow we consider a directed graph representing a network of pipes
- We choose one vertex as the source and a second vertex as a sink
- Each edge has a flow capacity that cannot be exceeded
- The problem is to maximise the flow between source of sink
- This can be used to model the flow of a fluid, parts in an assembly line, current in an electrical circuit or packets through a communication network

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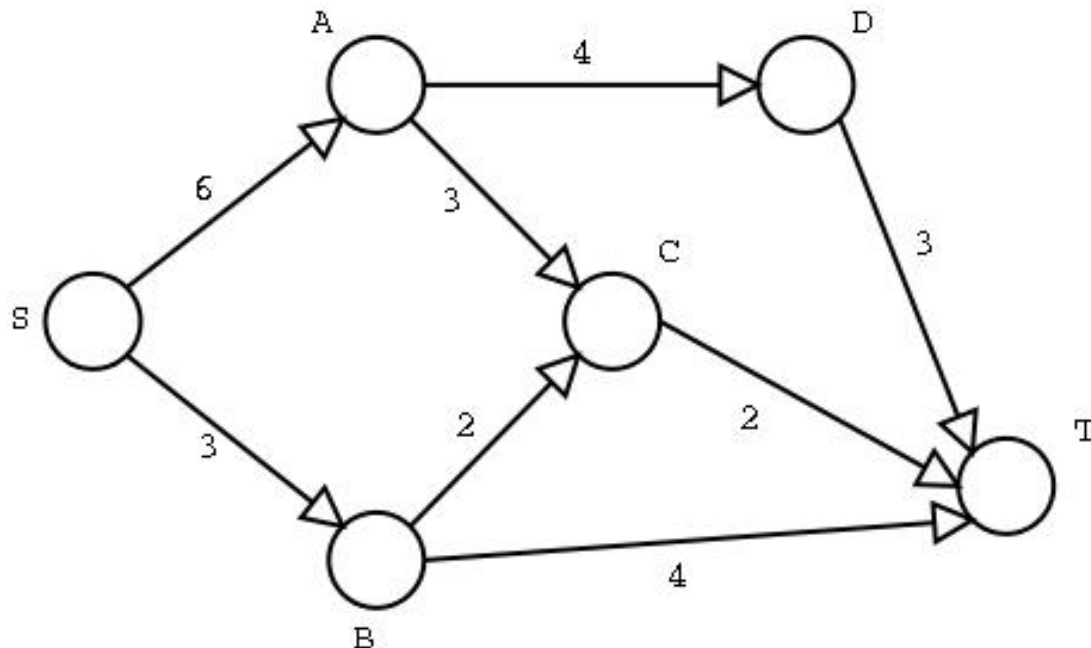
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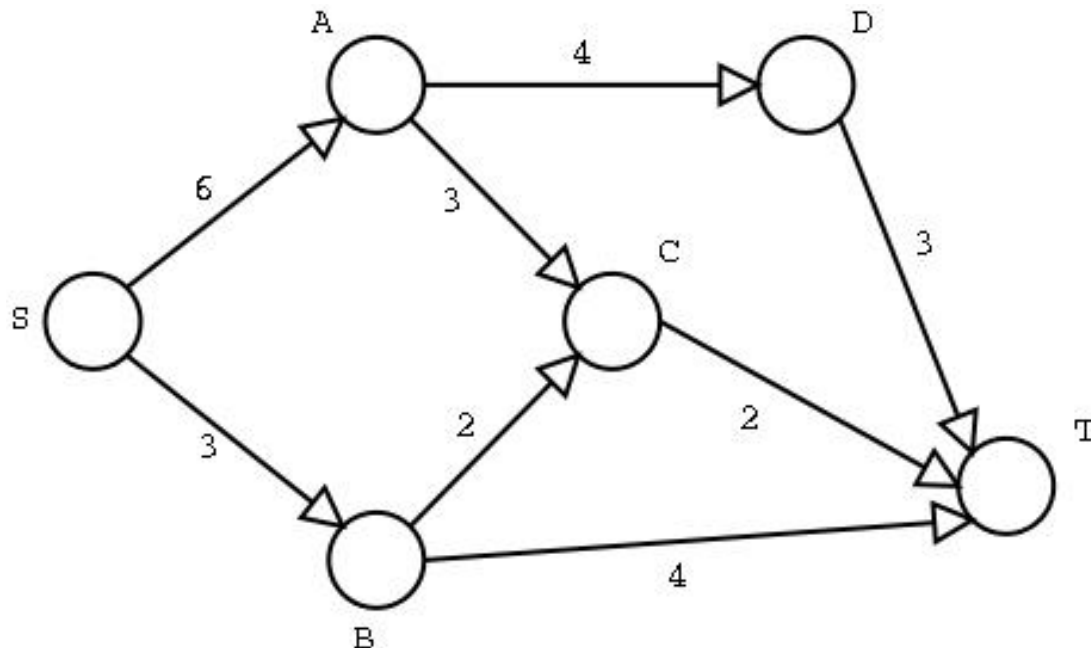
Example

- Consider a firm that has to ship haggis from Edinburgh to Southampton
- The shipping firm transports this in crates which it sends through intermediate cities
- The number of crates is limited by the size of the lorries it uses



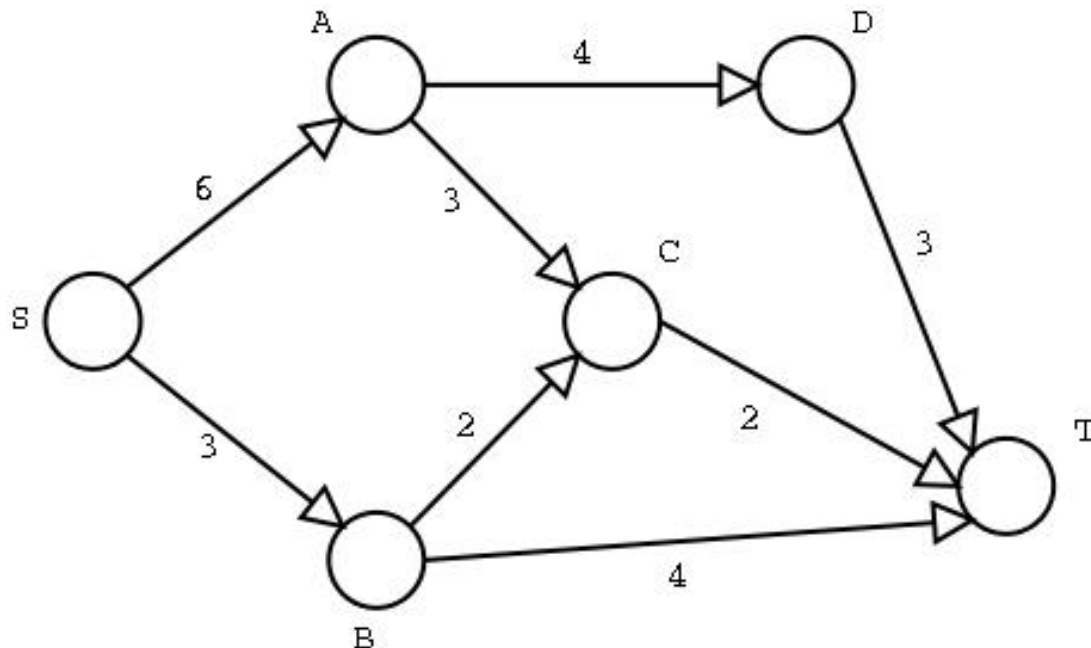
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Flow

- We are given a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where each edge has a capacity $c(i, j)$
- We define the flow from i to j as $f(i, j)$ with $0 \leq f(i, j) \leq c(i, j)$
- For all vertices except the source (s) and sink (t) we assume

$$\forall i \in \mathcal{V} / \{s, t\} \quad \sum_{j \in \mathcal{V} | (i, j) \in \mathcal{E}} f(i, j) = \sum_{j \in \mathcal{V} | (j, i) \in \mathcal{E}} f(j, i)$$

(i.e. no flow is lost from source to sink)

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Solving Maximum Flow

- As set up we have a linear objective function with linear constraints
- We can therefore solve this problem with a LP-solver
- (Note the solution will typically involve a fraction flow)
- However, this is such a classic problem with a distinctive structure that we can solve it more quickly with other algorithms
- The classic algorithm is the Ford-Fulkerson method with run time $O(|\mathcal{E}| \times f_{\max})$ where $f_{\max}$ is the maximum flow, although we won't cover this in the course

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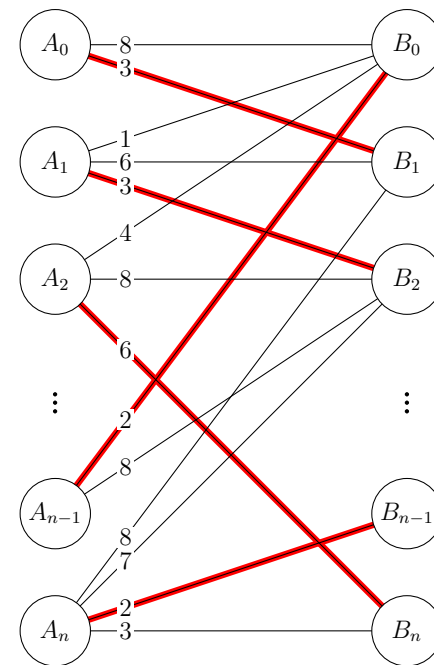
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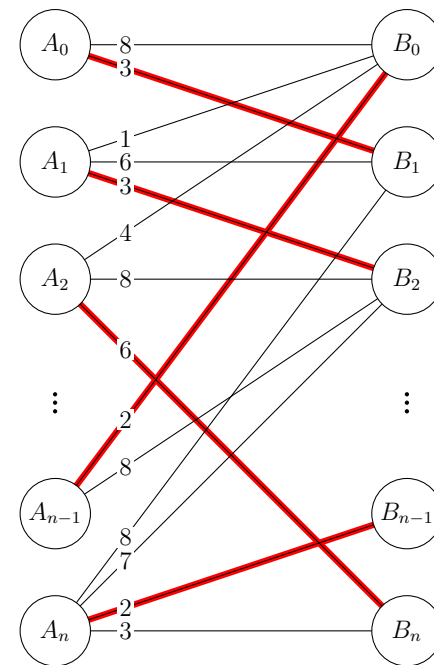
Linear Assignment

- We are given a set of n agents, $\mathcal{A}$, and n tasks, $\mathcal{T}$
- Each agent has a cost associated with performing a task $c(a, t)$
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- Consider a taxi firm with taxi's at 5 different locations and 5 requests to fulfil. The cost is the distance to the clients. Which taxi should go to which client?



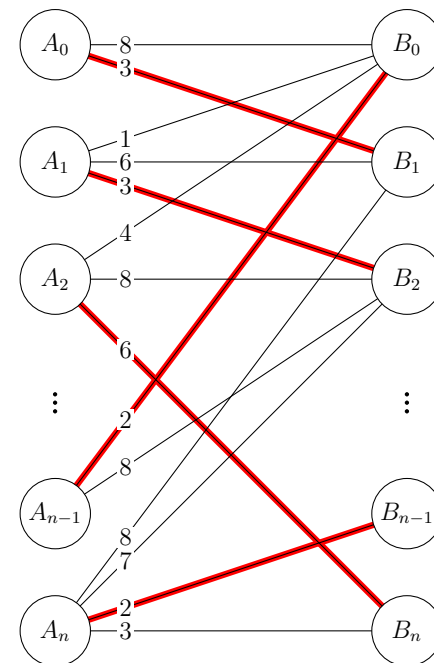
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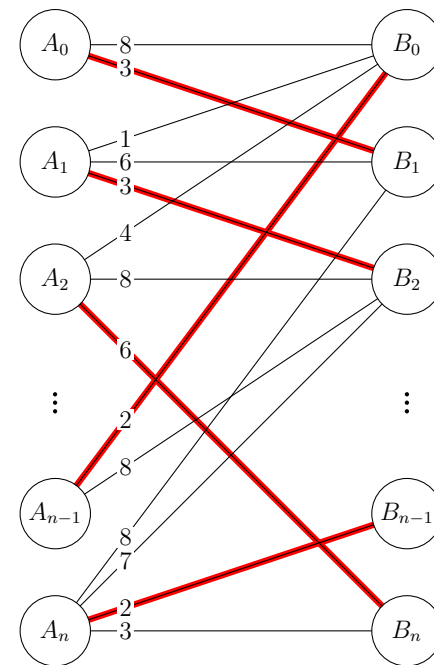
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LA as LP

- The linear assignment problem can be set as a linear programming problem

$$\min_x \sum_{a \in \mathcal{A}, t \in \mathcal{T}} c(a, t) x_{a, t}$$

subject to

$$\forall a \in \mathcal{A} \quad \sum_{t \in \mathcal{T}} x_{a, t} = 1$$

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$$\forall (a, t) \in (\mathcal{A}, \mathcal{T}) \quad x_{a, t} \geq 0$$

Hungarian Algorithm

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- Although it can be solved using a generic LP-solver this is not the most efficient algorithm
- The most efficient algorithm is the Hungarian algorithms
- This is rather complex (having once implemented it I can tell you from bitter experience it ain't easy)
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- We know a huge number of problems are solvable in polynomial time because they can be formulated as linear programs
- Linear programs occur sufficiently often that they are hugely important
- They aren't easy to solve, although standard simplex is not massively complex
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